BUSINESS INTELLIGENCE & ANALYTICS PROJECT

AdventureWorks Sales Performance Dashboard

Power BI · Multi-Page Interactive Report · 2024

$865K	$708.69K	542	847
Total Sub Sales	Total Line Sales	Products Tracked	Customers Analyzed

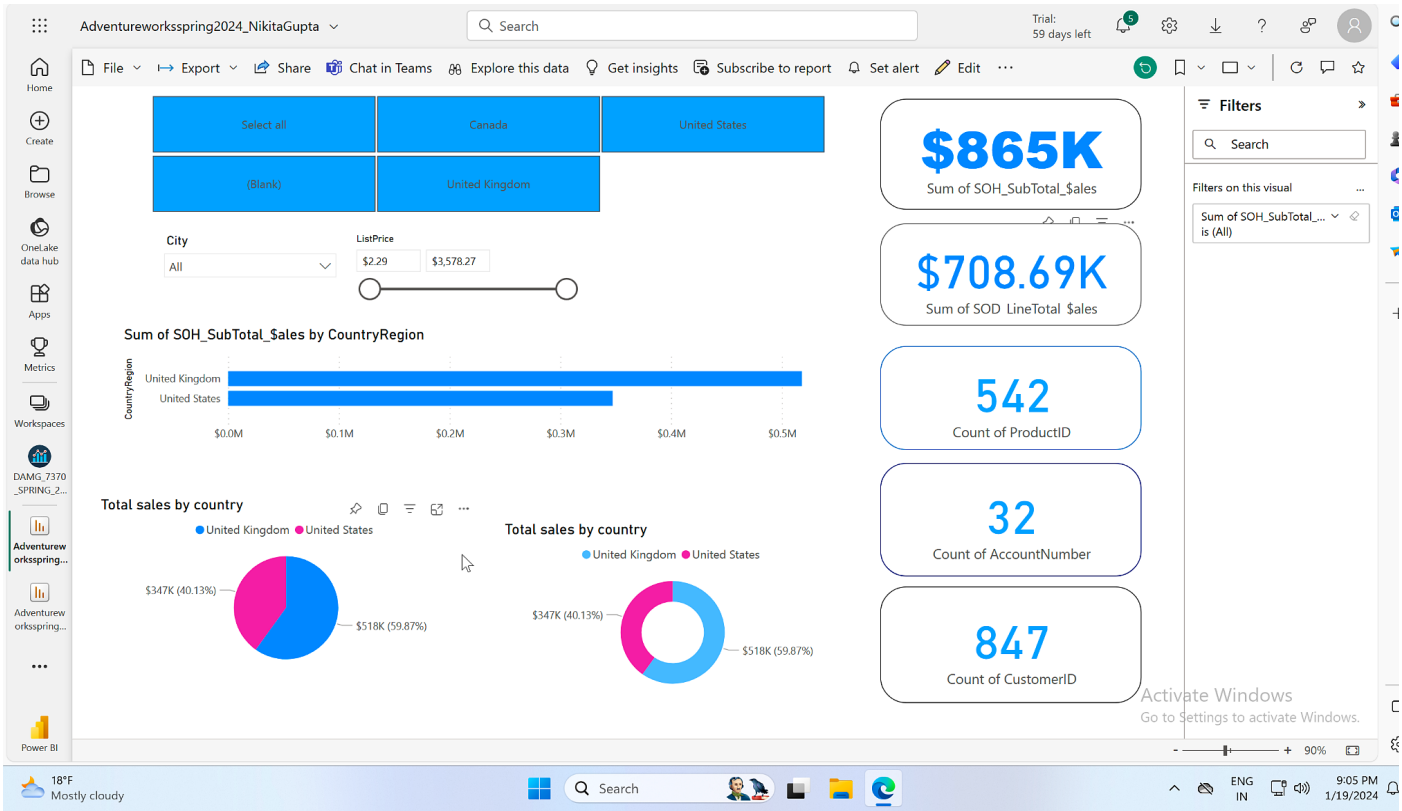

Fig 1 · Executive KPI Overview — Country-level sales snapshot

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Project Overview

The AdventureWorks Sales Performance Dashboard is a fully interactive, multi-page Power BI report built on the AdventureWorks dataset — a widely-used enterprise sales dataset covering global transactions, product lines, and customer accounts. The goal was to transform raw transactional data into actionable business intelligence that enables leadership to monitor revenue, compare regional performance, and identify high-value markets at a glance.

Project Objectives

- Build an executive-level KPI dashboard tracking total sales, product count, and customer volume.
- Enable geographic drill-down from Country level down to State/Province and City level.
- Visualize sales distribution using bar charts, treemaps, and interactive geographic maps.
- Design cross-filtering slicers for Country, State, and City dimensions across all pages.
- Deliver a 3-page report with distinct analytical perspectives per page.

Technology Stack

Tool / Technology	Purpose
Microsoft Power BI Desktop	Report design, DAX measures, and interactive visuals
AdventureWorks Dataset	Source transactional data covering sales, customers, and products
DAX (Data Analysis Expressions)	Calculated KPIs, aggregations, and dynamic measures
Power Query (M Language)	Data transformation, cleaning, and schema shaping
Power BI Service	Cloud publishing, sharing, and team collaboration

Dashboard Walkthrough

Report Page 1 — Executive KPI Overview

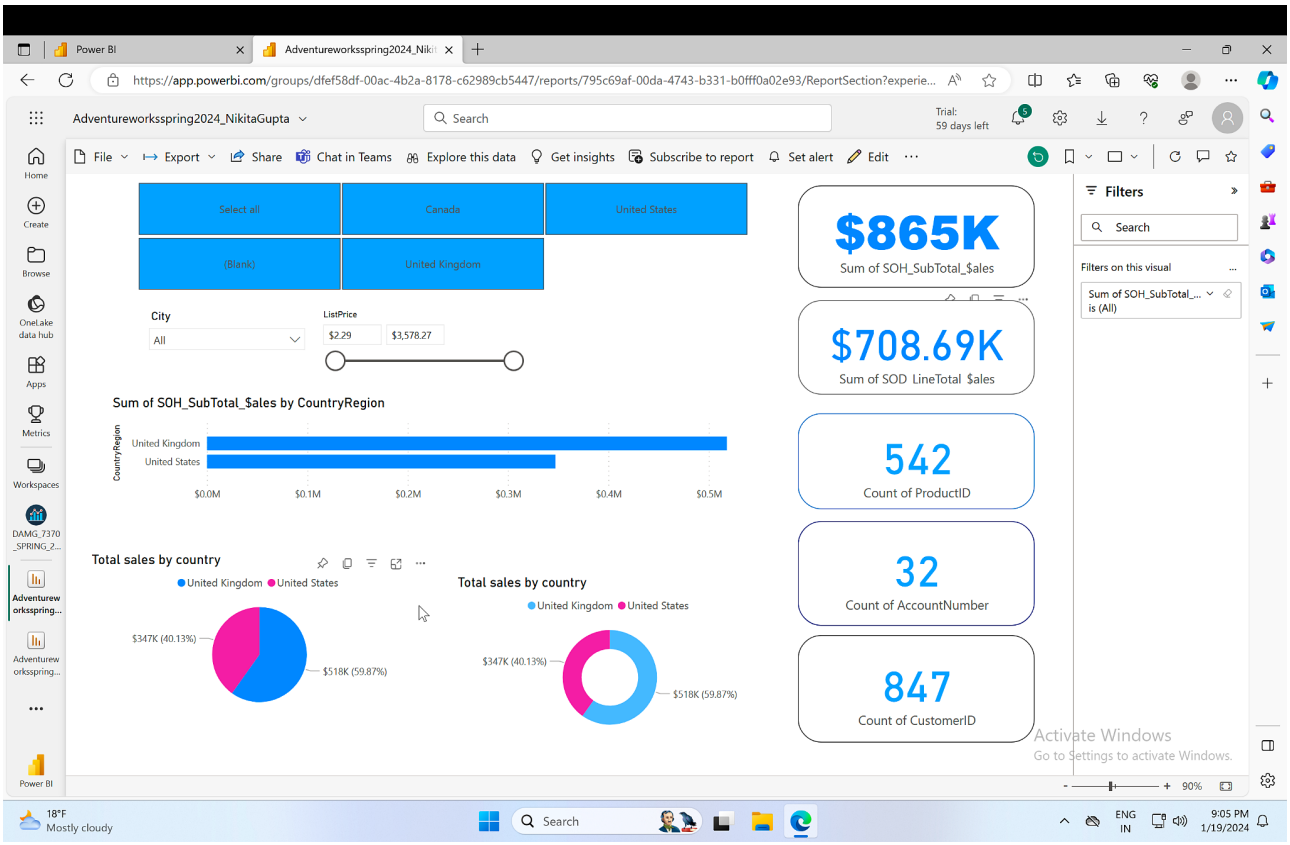

Fig 1 · Executive KPI Overview — Country-level sales with KPI cards and pie charts

The first page serves as the **executive summary view** of the AdventureWorks sales data. Four KPI cards are prominently displayed at the top right, giving leadership an immediate read on business health:

- **$865K** — Total SOH SubTotal Sales across all regions and time periods.
- **$708.69K** — Total SOH LineTotal Sales, reflecting net order line values.
- **542** — Distinct Product IDs sold, indicating portfolio breadth.
- **847** — Unique Customer IDs, showing the customer base served.

- A **horizontal bar chart** compares sales by Country/Region (United Kingdom vs United States).

- Two **pie charts** visualize proportional sales split — UK at ~40% and US at ~60%.

- Interactive **country slicers** allow filtering by France, Canada, United Kingdom, United States, and Mexico.

Report Page 2 — Regional Drill-Down Analysis

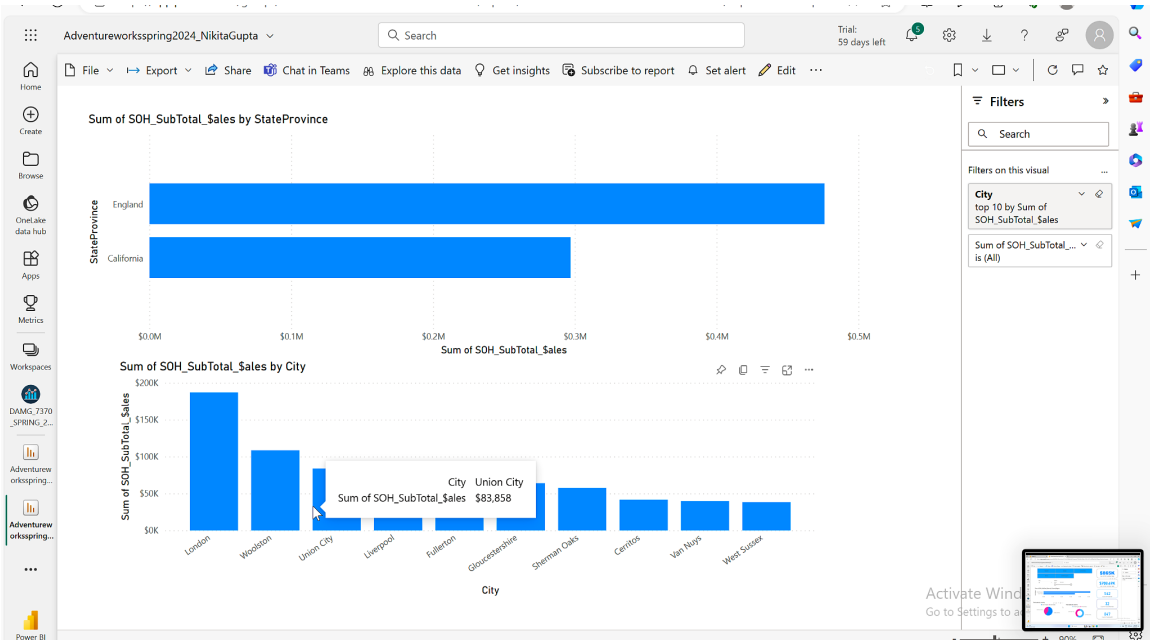

Fig 2 · State/Province and City-Level Sales Breakdown

The second page drills into **sub-national geographic performance**, allowing analysts to identify which states and cities are driving the most revenue:

• A **horizontal bar chart** (Sum of SOH_SubTotal_Sales by StateProvince) shows England and California as the top-performing regions.

• A **city-level bar chart** breaks down sales further — London, Woolston, and Union City among top cities, with Union City contributing $83,858.

• A **3-page navigation panel** is visible on the left, enabling seamless movement between report views.

• Filters on the visual allow analysts to focus on the **top 10 cities by sales** dynamically.

Report Page 3 — Geographic Map and Treemap Views

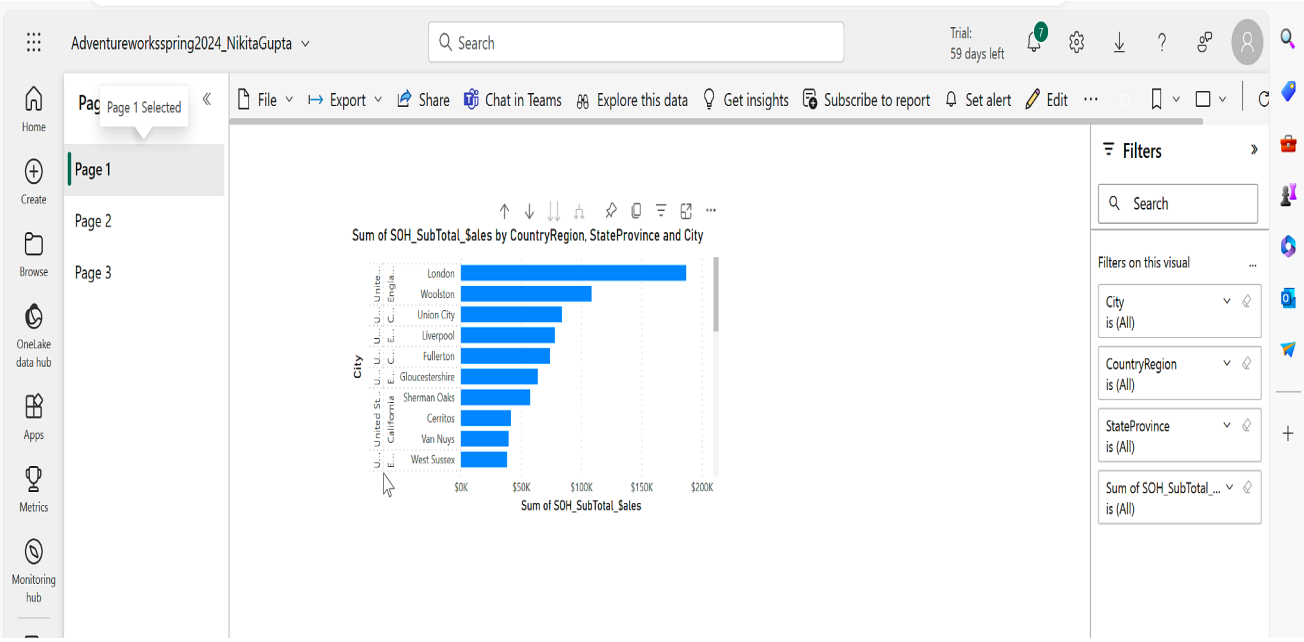

Fig 3 · Interactive Map and Treemap for Multi-Dimensional Sales Analysis

The third page introduces **spatial and hierarchical visualizations** for a richer analytical experience:

• An **interactive map visual** (Sum of SOH_SubTotal_Sales by StateProvince and CountryRegion) plots
sales bubbles geographically across the US and UK.

• A **treemap** shows city-level sales by relative size and color — London dominates the UK tile, while cities like Woolston, Liverpool, and Fullerton appear in the US cluster.

• Cross-filtering between Country, StateProvince, and City slicers makes the map and treemap fully dynamic and explorable.

• This page is optimized for **territory planning** and identifying underperforming regions visually.

Key Features & Technical Highlights

DAX-Powered KPI Cards	Custom DAX measures calculate total sales values dynamically, updating in real time based on active slicer selections across all pages.
Interactive Map Visual	Bing Maps integration displays geographic bubble charts by StateProvince and CountryRegion, enabling spatial sales analysis with no additional GIS tooling.
Treemap Visualization	Hierarchical treemap encodes both sales volume (size) and relative ranking (color) across cities, making it easy to spot top markets at a glance.
Cross-Page Filtering	Slicers for Country, StateProvince, and City are synchronized across all 3 pages, ensuring consistent filtering behavior throughout the entire report.
Dual Chart Comparison	Side-by-side pie charts and bar charts provide both proportional and absolute views of the same data, catering to different analytical preferences.
Multi-Page Report Structure	Organized into 3 focused pages: Executive Summary, Regional Drill-Down, and Geographic Analysis, following BI best practice for progressive disclosure.

Skills Demonstrated

Power BI Desktop	**DAX Measures**	**Data Modeling**	**Power Query**
KPI Dashboard Design	**Geographic Mapping**	**Cross-filtering**	**Treemap / Charts**

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